HAPPIEST HEALTH DIGITAL TWIN CHALLENGE · EXPERT JURY PREPARATION

T2D Glucose Digital Twin

A comprehensive, evidence-first technical explainer, scientific integrity review, and jury defense manual for Akshaya R and Nikhil VS.

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Default Branch: `release/candidate-clean`
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Live Workstation: `t2d-glucose-digital-twin.akshay.blitz.cloud`
Submission Video: `youtu.be/arTnv1yWy1E` (19:41)
Scope: Research Prototype Only — No Clinical Claims
Date: October 2026

> ⚠️ **Mandatory Scientific Integrity Rule for Presentation**
>
> All data presented during the jury defense must clearly distinguish between **synthetic development/validation findings** and **retrospective real-patient research**. Never claim clinical efficacy, diagnostic clearance, or superiority over simpler baselines where the evidence did not confirm it. If asked about an unmeasured metric or unexecuted study, say: *"We have not established that yet; our current evidence shows [verified result], and our next step would be [specific experiment]."*

1. The Project in 60 Seconds

What it is: A physiology-informed digital twin software workstation that forecasts continuous glucose monitor (CGM) readings two hours ahead and provides advance early warnings before postprandial glucose spikes (>180 mg/dL) occur in adults with type 2 diabetes and prediabetes.

Whose problem it is intended to address: Millions of patients and clinicians managing type 2 diabetes. Commercial CGMs only alert *after* glucose crosses 180 mg/dL (reactive alarms), when it is too late to prevent the excursion. Our system predicts the impending rise 30–60 minutes in advance, giving patients an actionable window for light activity (e.g. a 15-minute walk) or meal moderation.

What we built:

1. **Physiological Residual Engine:** Couples a mechanistic modified Bergman Minimal Model (solved via an Unscented Kalman Filter) with a deep Temporal Fusion Transformer (PR-TFT) that learns only the residual errors the physics cannot capture.
2. **Uncertainty & Spike Hazard:** Outputs 24 discrete 5-minute predictions (120-minute horizon) across 7 quantiles (P05 to P95) with conformal coverage adjustment, plus an isotonic-calibrated discrete spike-onset hazard.
3. **Interactive Clinical Workstation:** A Next.js web application featuring cohort triage, real-time telemetry replay, timeline anchor pinning, "What-If" meal/activity simulation, bilingual patient notes (English/Hindi), and official HL7 FHIR R4 JSON bundle export.
4. **Public Evidence & Governance Ledger:** Embedded into the live product, disclosing all development metrics, locked real-patient test results, external validation checks, failed gates, and known artefacts without

concealment.

What the Prototype Demonstrates

- Connected streaming replay of 249 served synthetic subjects.
- Live additive breakdown: Twin + Adjustment = Forecast.
- Real-time What-If scenario re-simulation (carbs & exercise).
- Automatic fallback to physics-only rollout if ML fails.
- Strict refusal of unmodelled medication interventions.

What it Does NOT Prove

- No clinical trials or proven HbA1c reductions.
- Not an automated medical device; cannot prescribe drugs.
- What-If is a simulated response, not a causal clinical effect.
- Real-patient early warning sensitivity is modest (6.9%).
- Synthetic data realism gate failed (AUROC 0.932).

2. The Healthcare Problem

Continuous Glucose Monitoring (CGM)

Continuous glucose monitors use a miniature subcutaneous enzymatic sensor to sample interstitial glucose every 5 minutes (288 readings/day). Interstitial fluid glucose lags blood glucose by 5 to 15 minutes due to physiological diffusion time between vascular capillaries and the interstitial matrix.

What an "Adverse Spike" Means

Under international clinical consensus, postprandial glucose should remain <180 mg/dL (10.0 mmol/L). In our preregistered evaluation protocol, an adverse spike is strictly defined as:

- Crossing from below 180 mg/dL to **≥180 mg/dL**.
- Preceded by a **rise of at least 50 mg/dL** within the prior 60 minutes.
- Persisting for **at least 3 consecutive 5-minute readings** (15 minutes), filtering out single-sample sensor noise and compression artefacts.

Why Advance Prediction is Valuable in Type 2 Diabetes

Unlike type 1 diabetes—where insulin boluses dominate glycemic control—type 2 diabetes management relies heavily on meal composition and muscle glucose uptake. Light physical exercise (e.g., a 20-minute post-meal walk) stimulates non-insulin-dependent GLUT4 transporter translocation, clearing glucose from plasma. However, this is effective primarily *before* or *during* the early postprandial rise. A 30- to 60-minute advance warning provides a practical, non-pharmacological intervention window.

> ⚠️ **Clinical Distinction: Prediction ≠ Prescription**
>
> A statistical forecast is a conditional probability distribution: $P(G_{t+h} \mid ext{history})$. Even when calibrated, it does not guarantee how a patient's metabolism will respond to an unmeasured behavioral change. In diabetes, unverified clinical recommendations carry severe risks of triggering acute hypoglycaemia (<70 mg/dL). Hence, our system strictly separates **forecasting** from **medical advice**.

3. What a Digital Twin Means in This Project

In this project, a "digital twin" is not a marketing buzzword. It is a mathematically defined computational state model that synchronizes static clinical priors with dynamic wearable observations:

- **Static EHR Context:** Age, sex, BMI, baseline HbA1c, eGFR, diabetes diagnosis duration, and prescribed medication classes.
- **Streaming Dynamic Inputs:** 5-minute CGM readings, heart rate, HRV (RMSSD), pedometer steps, sleep stages, and logged meal carbohydrates.
- **Hidden Physiological State Vector:** Solved recursively via an Unscented Kalman Filter tracking plasma glucose $G(t)$, remote insulin action $X(t)$, plasma insulin proxy $I(t)$, and gut glucose absorption compartments $Q_1(t)$ and $Q_2(t)$.

> **ℹ️ Cohort Distinction: Synthetic vs. Real Research**
>
> The live application workspace runs on a **synthetic cohort** of 249 served subjects (205 train, 44 validation) generated by an ODE simulator (`generate_health_data_v2`). The 44 synthetic test subjects are quarantined behind a server firewall. Retrospective research evaluations were executed separately offline on 161 real patients across open clinical datasets.

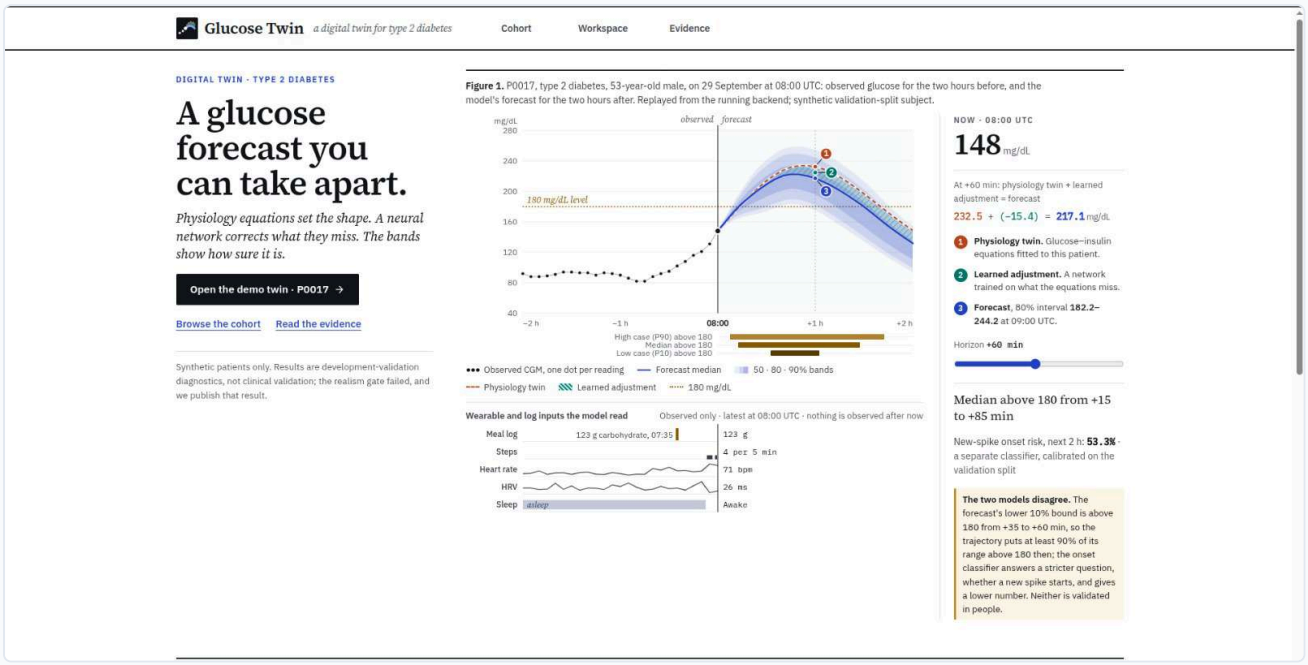
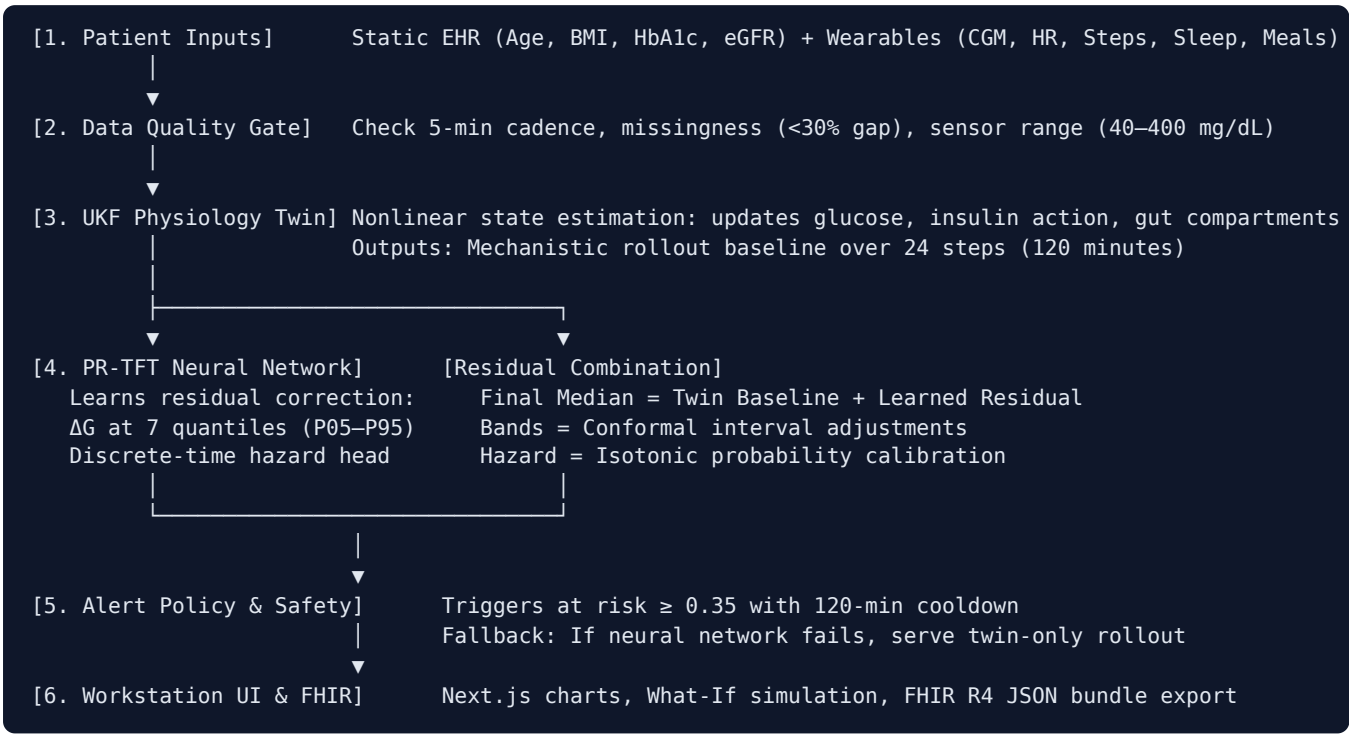


Figure 1: Current live application landing page showing patient P0017 (validation split, Type 2 diabetes) replayed from the running FastAPI backend at 08:00 UTC. Note the additive decomposition strip, wearable input timeline, and explicit model disagreement disclosure.

4. Exactly How Our System Works

The processing pipeline follows a causal 6-stage architecture where each stage has a clearly defined responsibility and fallback:



1. UKF Physiology Twin (Modified Bergman Minimal Model)

Pure deep learning models are prone to hallucinating physiologically impossible trajectories during sensor dropouts. To guarantee stability, our system anchors every forecast on a biophysical prior:

- **Bergman Minimal ODEs:** Model insulin action accelerating glucose disposal and gut compartments absorbing meal carbohydrates into blood.
- **Unscented Kalman Filter:** Uses deterministic sigma points to propagate mean and covariance through nonlinear differential equations without linearisation instability. As each 5-minute CGM reading arrives, it updates the patient's current insulin sensitivity and gut state.
- **Twin Rollout:** Projects forward 120 minutes assuming no unannounced future food intake, producing a smooth, biologically plausible trajectory.

2. PR-TFT Neural Network (Residual Learning)

The Minimal Model cannot capture dawn phenomenon, circadian cortisol variation, physical exercise recovery, or stress. The Physiological-Residual Temporal Fusion Transformer (PR-TFT) learns strictly the **residual error** ($\Delta G = G_{ ext{actual}} - G_{ ext{twin}}$):

- Trained on static EHR features, 24-hour historical wearable series, and time-of-day clock features.
- Outputs 7 quantiles ($P_{05}, P_{10}, P_{25}, P_{50}, P_{75}, P_{90}, P_{95}$) simultaneously across 24 steps.
- **Additive Combination:** $ ext{Final Median Forecast} = ext{Twin Median} + ext{Learned Residual Median}$. On-screen, this is visible as an explicit strip in mg/dL.

3. Conformal Coverage & Isotonic Calibration

- **Conformal Intervals:** Raw quantile widths are adjusted using empirical non-conformity scores calculated on validation data to achieve valid 80% empirical coverage.
- **Isotonic Hazard Calibration:** The discrete hazard output is mapped through non-parametric isotonic regression to ensure that an estimated 40% risk corresponds to a true 40% event rate.

4. Safety Fallback

If the neural network crashes, runs out of memory, or fails input validation, the backend automatically serves the UKF physiological rollout alone. The API sets `model.fallback_used = true`, and the workstation notifies the clinician.

5. What Judges See in the Product

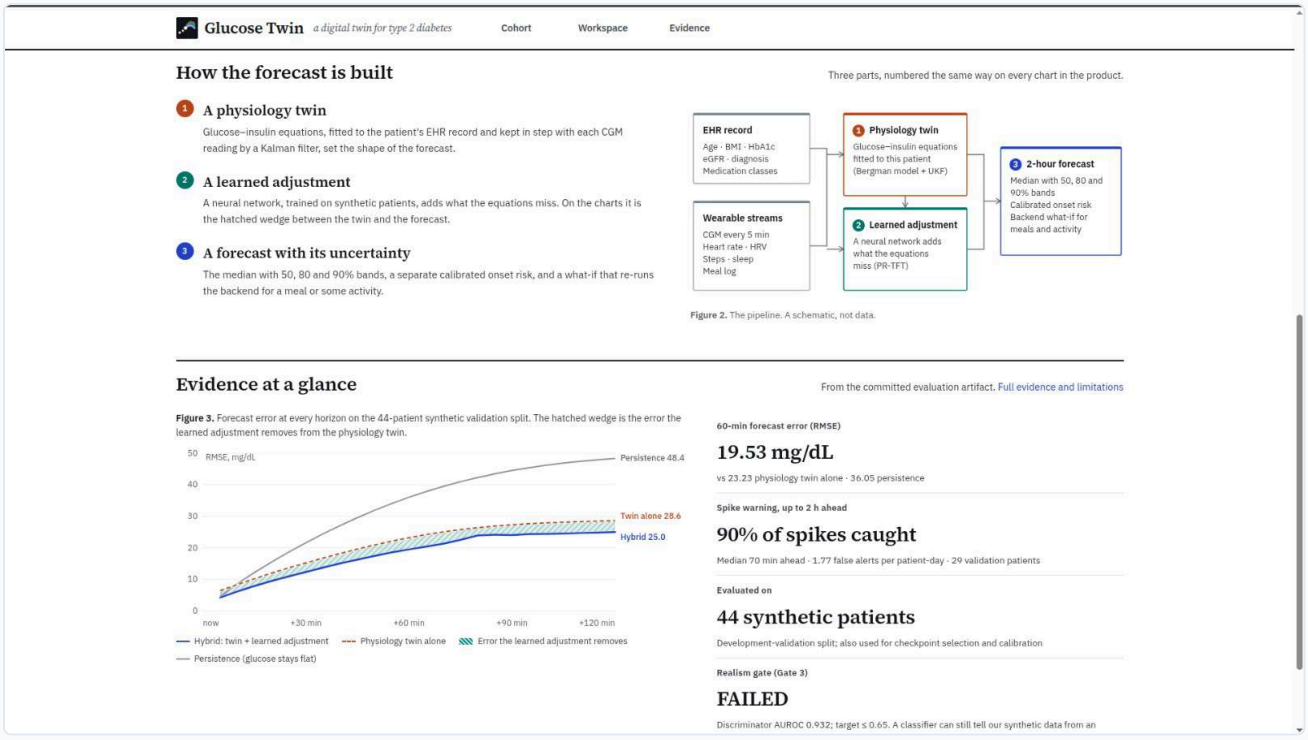


Figure 2: Pipeline architecture schematic and validation evidence at a glance on the live site, including the prominent disclosure that the Gate 3 synthetic realism audit failed (AUROC 0.932).

1. Patient Cohort Directory (`/cohort`)

Displays all **249 served synthetic subjects** (88 Healthy, 107 Prediabetes, 54 Type 2; 205 train, 44 val). Includes a **Ward Triage View** ranking patients at 08:00 UTC by their 2-hour spike risk (e.g. P0100 at 75%, P0283 at 61%, P0036 at 60%, P0017 at 53%). The 44 sealed test subjects are blocked.

2. Patient Workspace (`/patients/[id]`)

Observed CGM dots, forecast median, 50/80/90% uncertainty ribbons, physiological twin (dashed line), and learned adjustment (hatched teal wedge). Allows time-scrubbing and anchor pinning to evaluate historical points without future leakage.

3. What-If Scenario Exploration

Simulates adding a 60g carb meal in 30 minutes, 30 minutes of moderate activity, or skipping usual meals. Medication changes immediately return `SCENARIO_UNSUPPORTED` . Includes an authentic Indian Dish Selector (29 USDA FNDDS dishes).

4. Clinician Summary & FHIR Export

Generates an A4/Letter print export, a bilingual patient note (English/Hindi with disclaimer), and an official HL7 FHIR R4 JSON Bundle (LOINC 99504-3, UCUM mg/dL) validated against the official schema.

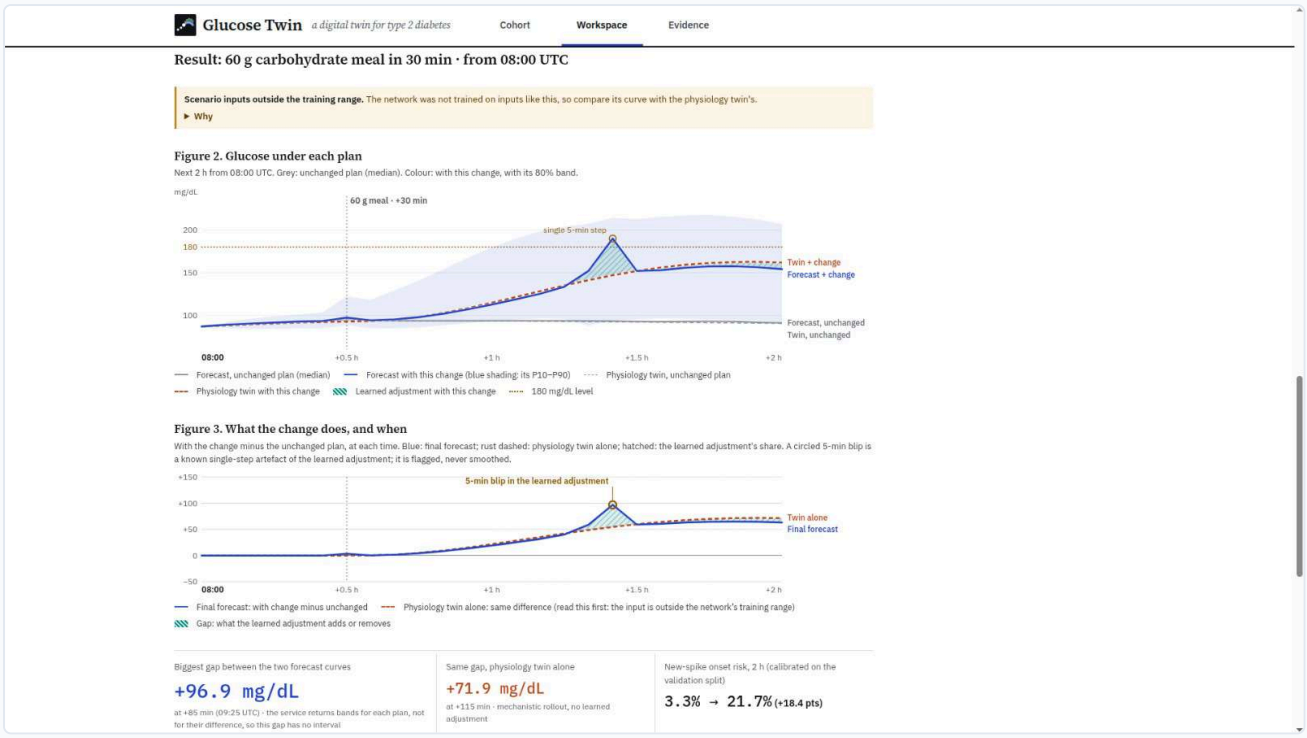


Figure 3: What-If simulation result for patient P0001 (60g carbohydrate meal in 30 minutes). Note Figure 2 (glucose under each plan) and Figure 3 (what the change does). The circled 5-minute blip in the learned adjustment is a known neural network artefact that is visibly flagged rather than hidden.

6. Model and Technical Architecture

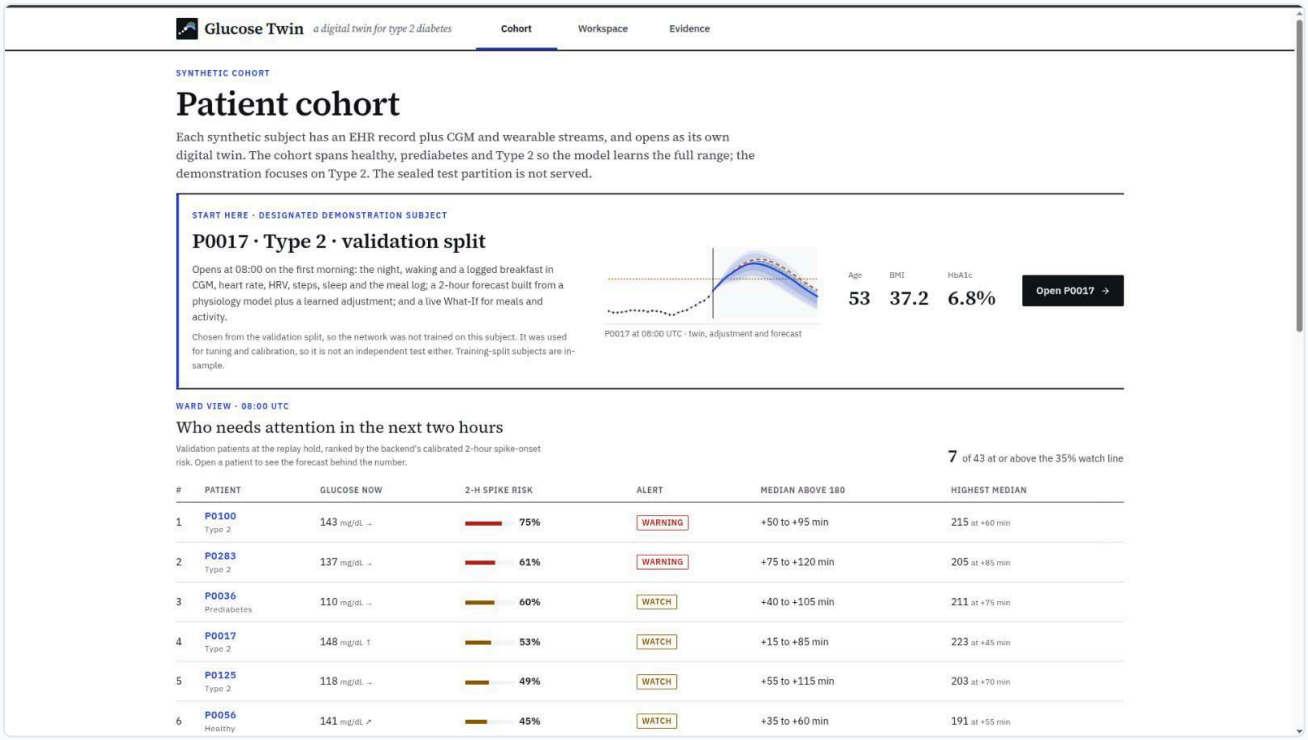


Figure 4: The Cohort Ward Triage view ranking served synthetic validation patients by calibrated 2-hour spike-onset risk at 08:00 UTC.

Production Software Stack

- **Frontend:** Next.js 15 (React 19, TypeScript, Tailwind CSS, Lucide icons), running as a standalone Node.js server.
- **Backend:** FastAPI (Python 3.12, PyTorch 2.x Lightning, NumPy, SciPy, Pydantic v2).
- **Physiological Module:** Modified Bergman ODE simulator + Unscented Kalman Filter (`backend/ukf_twin.py`).
- **Forecasting Module:** Physiological Residual Temporal Fusion Transformer (`backend/train_pr_tft.py`).
- **Deployment Infrastructure:** Single container running both services via `deploy/entrypoint.sh` on port 7860 under non-root `appuser` (UID 10001). Internal communication uses `http://127.0.0.1:8001`.

API Contract Specifications (contracts/v1/)

- `POST /v1/forecasts`: Accepts historical CGM, wearable streams, and optional scenario modifications. Returns 24 5-minute trajectory quantiles, conformal bounds, and calibrated spike hazard.
- `GET /v1/subjects/{id}/stream`: Server-Sent Events (SSE) streaming real-time CGM readings as if arriving live.
- `GET /v1/subjects/{id}/twin`: TwinCard reporting patient EHR priors and estimated physiological parameters.
- `GET /v1/subjects`: Enumerates served subject UUIDs and patient IDs (filters out sealed test partition).
- `GET /v1/health`: Service liveness, model version (`0.1.0`), simulator version, and active served subject count (`249`).

7. Dataset and Partition Story

A rigorous audit requires exact accountability for all cohorts, partitions, and denominators:

Cohort Phase	Dataset & Source	Total Size	Train / Val / Test Split	Status & Verification
Phase A: Synthetic	`generate_health_data_v2` (ODE simulator)	293 subjects	205 train / 44 val / 44 test	**Served in App: 249** (205 train + 44 val). 44 test subjects strictly sealed.
Phase B: Real Locked	CGMacros (PhysioNet 1.0.0, 5-min CGM)	45 patients	27 train / 9 val / 9 test	**Evaluated in Locked Test:** 9 test patients.
Phase B: Real Locked	BIG IDEAs (PhysioNet 1.1.3, 5-min CGM)	16 patients	9 train / 3 val / 4 test	**Evaluated in Locked Test:** 2 evaluable test patients (2 excluded for coverage).
Phase B: Real Locked	ShanghaiT2DM (Figshare v5, 15-min CGM)	100 patients	60 train / 20 val / 20 test	**Declared NOT ESTIMABLE:** 15-min cadence fails ≥70% 5-min coverage rule.
Phase C: External Check 1	Hall et al. 2018 (mostly normoglycaemic)	57 participants	Unused external	64 spike onsets across 1,104 patient-days. **Result: NOT CONFIRMED.**
Phase C: External Check 2	AZT1D (Type 1 diabetes on AID)	24 adults	Unused external	2,814 spike onsets across 988.6 patient-days. **Result: NOT MET.**
Phase C: External Check 2	Broll et al. (Type 2 diabetes)	5 patients	Unused external	104 spike onsets across 51.2 patient-days (descriptive only).
Phase D: Future Prereg.	AI-READI (FAIRhub, Dexcom G6)	~944 T2D patients	Test partition only	**FROZEN, UNEXECUTED:** Awaiting institutional data access sign-off.

> 📌 **Denominator Transparency**
>
> The headline real-patient test evaluation denominator is **11 held-out test patients** with native 5-minute CGM (9 from CGMacros, 2 from BIG IDEAs), drawn from a nominal pool of 161 participants. The 20 Shanghai test participants were honestly reported as *Not Estimable* because their 15-minute readings could not evaluate the 5-minute spike persistence rule without fabricated interpolation.

8. Results: What We Know

Synthetic Development Validation (44 Patients)

Source: `runs/pr_tft_ablation_arm_d/run_manifest.json` *&* `docs/evidence/validation_ceiling_results.json`

Forecast Horizon	PR-TFT Hybrid RMSE [95% CI]	Physics Twin Alone	Persistence Baseline	Relative Error Reduction vs. Physics
+30 min	**12.36 mg/dL** [11.89–12.84]	15.24 mg/dL	21.64 mg/dL	−18.9%
+60 min	**19.53 mg/dL** [18.56–20.48]	23.23 mg/dL	36.05 mg/dL	**−16.0%** (−45.8% vs. persistence)
+120 min	**25.03 mg/dL** [23.67–26.27]	28.64 mg/dL	48.36 mg/dL	−12.6%

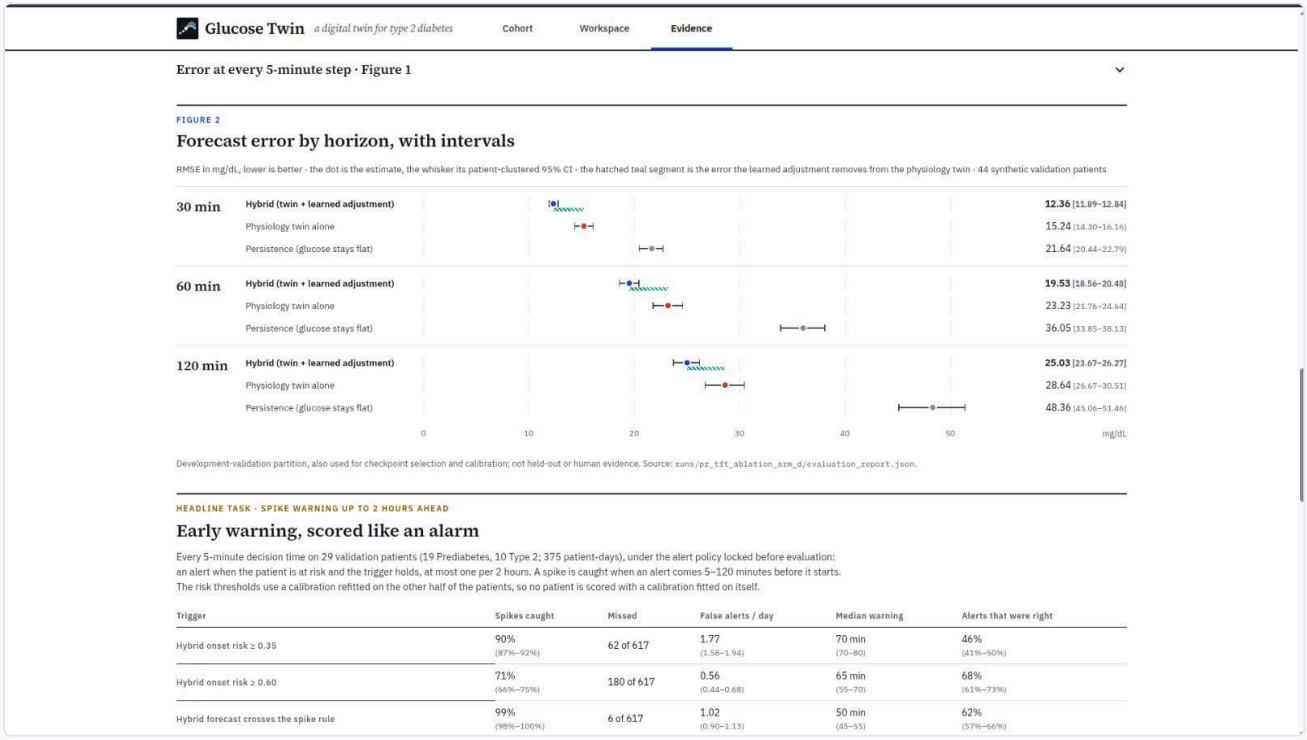


Figure 5: Forecast error by horizon with patient-clustered 95% confidence intervals, and alarm-scored early warning performance on synthetic validation data.

Locked Real-Patient Evaluation (11 Held-Out Test Patients)

Source: `docs/evidence/real_data/results.json` *(Point Estimates) and* `RESULTS.md` *(Bootstrap Means)*

Model	60-min RMSE (mg/dL) [95% CI]	Sensitivity ≥30m [95% CI]	Alert Precision	False Alerts / Day [95% CI]	Median Lead Time
Persistence Baseline	26.27 [20.85–32.22]	0.0% [—]	—	0.00 [—]	0.0 min
Linear Trend (30 min)	49.06 [42.10–57.40]	44.4% [36.8–56.1]	16.4%	**3.84** [3.16–4.40] (Over budget)	25.0 min
Reduced Twin (UKF)	24.73 [20.43–29.68]	0.0% [0.0–0.0]	30.8%	0.10 [0.00–0.31]	5.0 min
Masked Hybrid (Point Est.)	**23.11** [18.75–28.12]	**6.9%** [0.0–13.9]	**65.5%**	**0.11** [0.02–0.23]	**10.0 min** [5.0–15.0]*
Masked Hybrid (Bootstrap Mean)	*23.04 [18.7–28.1]*	*6.0% [0.0–13.9]*	*65.5%*	*0.11 [0.02–0.23]*	*10.0 min [5.0–15.0]**

**The median lead-time interval [5.0–15.0] is a fixed reporting band (±5 min), not a bootstrap confidence interval.*

🔍 Point Estimates vs. Bootstrap Means Reconciliation

As established in `docs/submission/LOCKED_RESULTS_RECONCILIATION.md`, `results.json` reports the exact **point estimate** on the full test set (RMSE 23.11 mg/dL, sensitivity 6.9%), while the narrative table in `RESULTS.md` prints the **mean of 2,000 bootstrap resamples** (RMSE 23.04 mg/dL, sensitivity 6.0%). Both originate from the exact same execution run. We quote point estimates as our headline standard.

9. Negative Results, Failed Gates, and Known Artefacts

We present our negative results with absolute transparency. This is our greatest scientific strength:

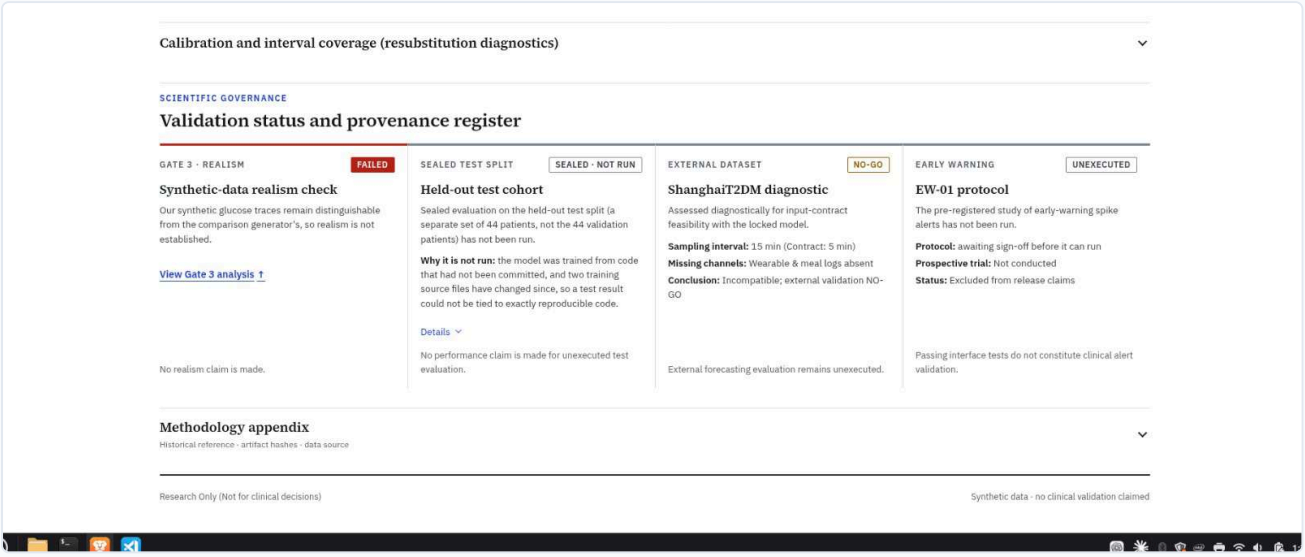


Figure 6: The live validation status and provenance register from the /evidence page, publicly showing Gate 3 FAILED, Sealed test split NOT RUN, Shanghai diagnostic NO-GO, and EW-01 UNEXECUTED.

1. Gate 3 Realism Audit: FAILED

A 40-feature random forest discriminator attempted to distinguish our synthetic CGM data from an independent mock-real generator. Target was AUROC ≤ 0.65. The classifier scored **AUROC = 0.932** (FAILED). Distinguishing feature: `jerk_sd` (synthetic ODEs were too smooth compared to real sensors). Disclosed on landing page and evidence page.

2. Low Real Early-Warning Sensitivity

On held-out real patients, the hybrid model caught only **6.9%** of spikes ≥30 minutes in advance. Because retrospective meal logs lacked verified timestamps, meal inputs were masked to zero; the conservative probability threshold (0.35) stayed silent until the rise had already begun to avoid false alarms.

3. External Checks: NOT CONFIRMED / NOT MET

On Hall et al. (57 participants), the routine model improved early warning (+7.8%), but the 95% CI was [0.0, +13.2], missing the pre-registered requirement of lower bound > 0. On AZT1D (24 T1D adults), the learned baseline (−2.5%) lost to a simple linear trend. Fixed probability thresholds fail to transfer across populations without local recalibration.

4. Single-Step 20 mg/dL Median Jumps

In ~19.8% of synthetic validation forecasts, the neural network median jumps >20 mg/dL in a single 5-minute step (an unphysical ML artefact). A pre-registered 3-point median filter cuts this to 2.3%. However, we refused to silently patch the weights; instead, the live UI flags these jumps with a circled warning indicator.

10. Privacy, Partition Protection, and Safety Governance

Data Partition Firewall

- **Served Cohort:** Strictly restricted to the 205 training and 44 validation synthetic subjects.
- **Sealed Test Cohort:** The 44 synthetic test subjects are quarantined behind `SEALED_SPLITS = frozenset({"test"})` in `backend/serve_pr_tft.py`. Querying any test subject returns HTTP 404.
- **Zero Real Patient Data Served:** Real patient records from PhysioNet and Figshare remain exclusively in offline research directories and are never bundled into Docker or exposed publicly.

Clinical Safety Safeguards

- **Research Disclaimer:** Prominently displayed across all application views and API metadata: `"research_only_not_for_clinical_decisions"`.
- **Medication Refusal:** Scenario requests attempting to adjust insulin or medications are rejected with `422 Unprocessable Entity` (`SCENARIO_UNSUPPORTED`) because the twin lacks validated pharmacokinetic drug terms.
- **FHIR R4 HTEST Tagging:** Exported JSON bundles carry the official `HTEST` (Healthcare Test Data) security tag to prevent accidental ingestion into production hospital EHR systems.

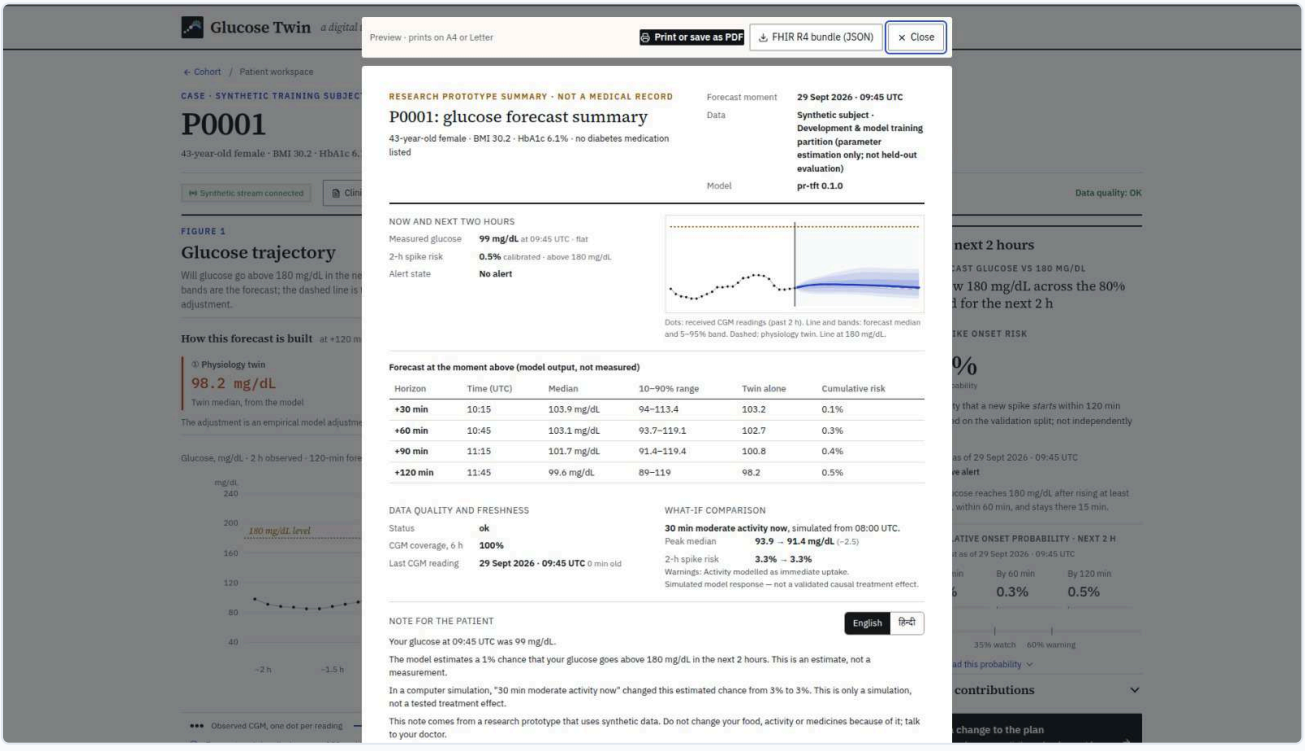


Figure 7: The Clinician Summary modal offering A4/Letter PDF export, FHIR R4 JSON Bundle export, data quality tracking, and a bilingual patient note (English/Hindi).

11. What We Would Do Next (Research Roadmap)

Our future research roadmap is grounded directly in the verified limitations identified during evaluation:

1. **Execute AI-READI Confirmation:** Once data access on FAIRhub is formally approved, execute the frozen, unedited `aireadi_evaluate.py` harness against ~944 type 2 diabetes participants wearing Dexcom G6 CGMs to test the C2 risk-gated trend policy on a massive, untouched real cohort.
2. **Dynamic Patient-Specific Operating Point:** Implement adaptive thresholding where alert thresholds dynamically scale with an individual's historical spike rate, resolving cross-population calibration drift.
3. **Unannounced Meal Deconvolution:** Build an analytical deconvolution estimator to infer carbohydrate influx directly from early CGM inflections, eliminating reliance on retrospective manual meal logs.
4. **Physiological Smoothness Regularization:** Retrain PR-TFT with a second-derivative curvature penalty ($\mathcal{L}_{ ext{smooth}} = \| abla^2 \Delta G\|^2$) to permanently prevent single-step 20 mg/dL jumps without post-hoc filtering.
5. **Prospective Clinical Usability Pilot:** Collaborate with an endocrinology clinic to observe how clinicians interact with forward-looking trajectory bands and bilingual summaries during routine outpatient reviews.

12. Glossary of Key Terms

- **CGM (Continuous Glucose Monitor):** Wearable subcutaneous biosensor sampling interstitial fluid glucose every 5 minutes.
- **Interstitial Lag:** 5–15 minute physiological delay between blood glucose and interstitial sensor readings.
- **Adverse Spike:** A rise of ≥50 mg/dL within 60 minutes crossing 180 mg/dL and persisting for ≥15 minutes.
- **Bergman Minimal Model:** Classic physiological two-compartment ODE system describing insulin-dependent glucose clearance.
- **Unscented Kalman Filter (UKF):** Recursive nonlinear state estimator that tracks internal metabolic states from noisy sensor inputs.
- **PR-TFT:** Physiological-Residual Temporal Fusion Transformer; deep attention model trained strictly to predict residual error over the physics twin.
- **Conformal Prediction:** Distribution-free calibration technique guaranteeing empirical coverage bounds (e.g. 80%).
- **1-to-1 Alert Matching:** Evaluation standard where one advance alert can credit at most one subsequent spike, preventing multi-event overcounting.
- **Refractory Cooldown:** Mandatory 120-minute alert suppression period following a trigger to eliminate alert fatigue.
- **Lead Time:** Minutes elapsed between an advance alert and the glucose trajectory crossing 180 mg/dL.
- **HL7 FHIR R4:** Fast Healthcare Interoperability Resources v4; standard JSON schema for healthcare data exchange.

13. One-Page Quick Reference Summary

Core Statistics to Quote

- **Synthetic Val 60m RMSE:** 19.53 mg/dL (−16.0% vs twin alone 23.23; −45.8% vs persistence 36.05).
- **Real Held-Out Test 60m RMSE:** 23.11 mg/dL [18.75–28.12] (11 held-out test patients).
- **Real Early Sensitivity (≥30m):** 6.9% point est. / 6.0% bootstrap mean [0.0–13.9%].
- **Real False Alerts:** 0.11 / patient-day [0.02–0.23] (≤1.0 budget).
- **Real Precision:** 65.5% (vs linear trend 16.4%).
- **Real Median Lead Time:** 10.0 min [5–15 min reporting band].
- **Gate 3 Realism Audit:** FAILED (AUROC 0.932 vs ≤0.65).
- **Served Deployed Cohort:** 249 subjects (44 test sealed).

Key Repository Locations

- `backend/serve_pr_tft.py` : FastAPI server & sealed partition firewall.
- `backend/ukf_twin.py` : Bergman ODEs & UKF filter.
- `backend/train_pr_tft.py` : PR-TFT neural network.
- `docs/evidence/real_data/` : Preregistration, results.json & RESULTS.md.
- `docs/submission/LOCKED_RESULTS_RECONCILIATION.md` : Metric reconciliation.
- `src/app/evidence/page.tsx` : Embedded evidence UI.
- `src/lib/fhir.ts` : Official HL7 FHIR R4 bundle builder.